Course Title: Discrete Mathematics

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Report Paper of Mini-Project

Group: 02

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Introduction:

This report explores the random generation of undirected graphs using C, represented by the adjacency matrix, for n = 1000, 2000, 3000, 4000 and 5000 vertices. It examines the number of edges, vertex degree, computation time, and verifies the handshaking theorem, resulting in a polynomial trendline analysis to determine the time complexity.

Program Output:

For n = 1000

```
D:\Untitled1.exe

Enter Number of Vertices: 1000
Number of Edges in 1000 Vertices graph is : 250202
Number of Degree in 1000 Vertices graph is : 500404
2|E| = Total degree, So handshaking logic holds
When n = 1000 Computational time is (in ms): 25.000000
Process returned 0 (0x0)   execution time : 3.948 s
Press any key to continue.
```

For n = 2000

D:\Untitled1.exe

```
Enter Number of Vertices: 2000
Number of Edges in 2000 Vertices graph is : 999534
Number of Degree in 2000 Vertices graph is : 1999068
2|E| = Total degree, So handshaking logic holds
When n = 2000 Computational time is (in ms): 72.000000
Process returned 0 (0x0)   execution time : 3.431 s
Press any key to continue.
```

For n = 3000

D:\Untitled1.exe

```
Enter Number of Vertices: 3000
Number of Edges in 3000 Vertices graph is : 2250670
Number of Degree in 3000 Vertices graph is : 4501340
2|E| = Total degree, So handshaking logic holds
When n = 3000 Computational time is (in ms): 165.000000
Process returned 0 (0x0)   execution time : 4.562 s
Press any key to continue.
```

For n = 4000

```
D:\Untitled1.exe

Enter Number of Vertices: 4000
Number of Edges in 4000 Vertices graph is : 4001060
Number of Degree in 4000 Vertices graph is : 8002120
2|E| = Total degree, So handshaking logic holds
When n = 4000 Computational time is (in ms): 242.000000
Process returned 0 (0x0)   execution time : 4.332 s
Press any key to continue.
```

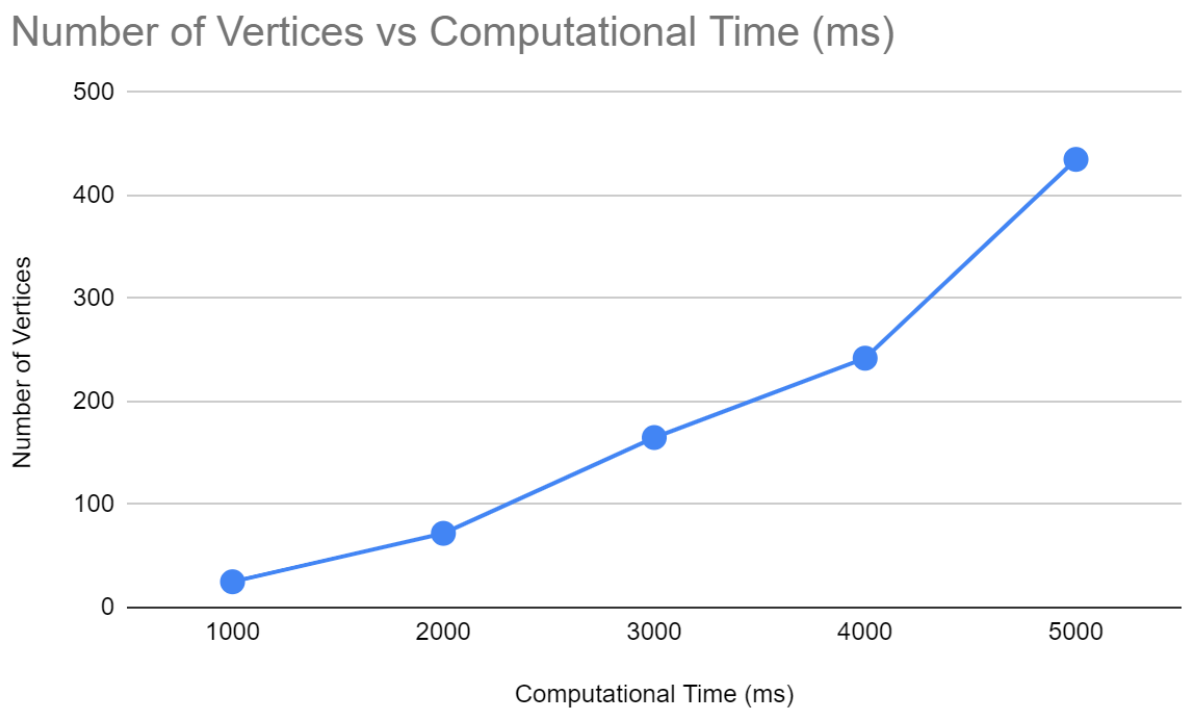
For n = 5000

```
D:\Untitled1.exe

Enter Number of Vertices: 5000
Number of Edges in 5000 Vertices graph is : 6251461
Number of Degree in 5000 Vertices graph is : 12502922
2|E| = Total degree, So handshaking logic holds
When n = 5000 Computational time is (in ms): 435.000000
Process returned 0 (0x0)   execution time : 4.927 s
Press any key to continue.
```

Graph showing computational time vs n:

Computational Time (ms)	Number of Vertices
25	1000
72	2000
165	3000
242	4000
435	5000



Graph: This graph shows the relationship between the calculation time and the variable n. It indicates that as the value of n increases, the time required for calculations also increases.

Conclusion: This project demonstrated that the computational time for analyzing network connections using an adjacency matrix grows quadratically

with the number of vertices ($O(n2)O(n2)$). The empirical data confirmed this trend, showing increasing computation times as the vertex count increased from 1000 to 5000. While the current approach is feasible for moderately large graphs, optimization is needed for very large graphs. Future work could focus on improving efficiency and exploring additional graph properties.